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John Christopher Lentfer Kristiansen
202005794

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Supervisor: Fabrice Baudoin

Institut For Matematik
Aarhus Universitet

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Chapter 1

We gotta start somewhere

In this thesis we will be studying the Vicsek set, so therefore we will in this section start by constructing the set and proving some basic properties of the Vicsek set. While it is possible to define the Vicsek set as a non-empty compact self-similar set, we will first define and prove some basic properties of self-similar sets.

1.1 Iterated Function System

The iterated functions systems (IFS) gives us a way to work with self-similarly sets. Another reason for constructing the IFS is that we are able to prove that the Vicsek set actually exists.

Definition 1.1: A finite family of functions $\{\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_m\}$ with $m \geq 2$ is called an *iterated function system* (or IFS) if exists $c_i \in (0, 1)$ such that all $\mathcal{S}_i$ are *contractions*. Which is exactly

$$\left|\mathcal{S}_i(x) - \mathcal{S}_i(y)\right| \leq c_i|x - y|\,.$$

Further we call a non-empty compact subset K of X an *invariant set* for the IFS if

$$K = \bigcup_{i=1}^{m} \mathcal{S}_i(K) \tag{1.1}$$

The next question will then be if there for every IFS exists an invariant set for the IFS? We shall actually prove that this is indeed true, but before we begin the proof that a unique invariant set exists we will need to define the Hausdorff metric.

Definition 1.2: Define $A_\delta = \left\{x \in \mathcal{A} : \exists a \in A, |x - a| < \delta\right\}$ The *Hausdorff metric* is given by

$$d_H(X, Y) = \inf\left\{\delta : X \subset Y_\delta \text{ and } Y \subset X_\delta\right\}$$

The proof that d_H metric can be found in [Hen99].

Theorem 1.3: Consider the IFS given by $\{\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_m\}$, then there is a unique invariante set K for the IFS.

Moreover, if we define a transformation S, on the class $\mathcal{A}$ of non-empty compact sets, by

$$S(E) = \bigcup_{i=1}^{m} \mathcal{S}_i(E), \quad \text{for } E \in \mathcal{A}$$

and write S^k to be the$k$'th iterate of S, so $S^0 = E$ and $S^k(E) = S(S^{k-1}(E))$ for $k \geq 1$. Then

$$F = \bigcap_{i=0}^{\infty} S^k(E)$$

for every set $E \in \mathcal{A}$ such that $S_i(E) \subset E$ for all i.

Proof. Let E be any set in $\mathcal{A}$ such that $S_i(E) \subset E$ for all i. By induction we find that $S^k(E) \subset S^{k-1}(E)$. Since $S^0 = E \supset S(E) = S^1(E)$, then $S^{k+1}(E) = S(S^k(E)) \subset S(S^{K-1}(E)) = S^k(E)$ by the induction assumption that $S^k \subset S^{k-1}$. So $S^k(E)$ is a decreasing sequence of non-empty compact sets, in which case it must have a non.empty compact intersection $K = \bigcap_{k\geq 0} S^k(E)$. Since $S^k(E)$ is decreasing, then $S(K) = K$ fulfilling (1.1).

Now moving on to uniqueness, We will need the following inequality

$$d_H(S(A), S(B)) = d_H\left(\bigcup_{i=1}^{m} \mathcal{S}_i(A), \bigcup_{i=1}^{m} \mathcal{S}_i(B)\right) \leq \max_{1\leq i\leq m} d_H(\mathcal{S}_i(A), \mathcal{S}_i(B)) \leq \max_{1\leq i\leq m} c_i \cdot d_H(A, B). \quad (1.2)$$

So assume A and B are both invariant sets, then $S(A) = A$ and $S(B) = B$, but that gives us that

$$d_H(A, B) = d_H(\mathcal{S}_i(A), \mathcal{S}_i(B)) \leq \max_{1\leq i\leq m} c_i \cdot d_H(A, B)$$

and since $0 < \max_{1\leq i\leq m} c_i < 1$, then $d_H(A, B) = 0$ so $A = B$ and it is unique.

Lastly we will prove (1.2). The first inequality comes by choosing $\delta = \max_{1\leq i\leq m} d_H(\mathcal{S}_i(A), \mathcal{S}_i(B))$ since that means that $\mathcal{S}_i(B) \subset (\mathcal{S}_i(A))_\delta$ for all $0 \leq i \leq m$, thus it will definitely hold for $\bigcup_{i=1}^m \mathcal{S}_i(B) \subset \left(\bigcup_{i=1}^m \mathcal{S}_i(B)\right)_\delta$ and vice versa. The second inequality comes from letting $\delta = d(A, B)$ then for every $a \in A$ we can choose some $b \in B$ such that $|a - b| \leq \delta$ in which case $\left|\mathcal{S}_i(a) - \mathcal{S}_i(b)\right| \leq c_i\delta$ and therefore $\mathcal{S}_i(A) \subset \left(\mathcal{S}_i(B)\right)_{c_i\delta}$. □

In the end we will mention a few consequences of of theorem 1.3. The theorem gives us a way to create a (not necessarily unique) sequence $(i_1, i_2, \ldots)$ such that each $x \in F$ we can write

$$\{x\} = \left\{x_{i_1,i_2,\ldots}\right\} = \bigcap_{k\geq 0} K_{i_1,\ldots i_k}.$$

This is possible since S^k is decreasing as k increases. This further lets us describe K as

$$K = \bigcup_{\mathcal{I}} \left\{x_{i_1,i_2,\ldots}\right\}. \quad (1.3)$$

1.2 Construction the Vicsek set

Let $V_0 = \{p_1, p_2, p_3, p_4, p_5\}$ be the set of vertices in a plus with a middle point in $\mathbb{R}^2$. Then define

$$\mathcal{S}_i(z) = \frac{1}{3}(z - p_i) + p_i, \quad \text{for } i = 1, 2, \ldots 5.$$

$\{\mathcal{S}_1, \ldots, \mathcal{S}_5\}$ is indeed a IFS, since for every $x, y \in \mathbb{R}$

$$\left|\mathcal{S}_i(x) - \mathcal{S}_i(x)\right| = \left|\frac{1}{3}(x - p_i) + p_i - \left(\frac{1}{3}(y - p_i) + p_i\right)\right| = \frac{1}{3}|x - y|.$$

Thus by theorem 1.3 we define the Vicsek set K is as the unique non-empty compact subset in $\mathbb{R}^2$ such that

$$K = \bigcup_{i=1}^{5} \mathcal{S}_i(K).$$

We note that V_0 is the boundary of K.

Since we are going studying the convergence of discretely defined operators to smooth operators we can define a discrete Vicsek set in the follow way. We start by define a sequence of sets $(V_m)_{m\geq 0}$ inductively:

$$V_{m+1} = \bigcup_{i=1}^{5} \mathcal{S}_i(V_m)$$

this gives us a sequence of Vicsek set graphs $(G_m)_{m\geq 0}$ whose vertices are $(V_m)_{m\geq 0}$. Further we will use the notation of $V_* = \cup_{m\geq 0} V_m$ and for any $p \in V_m$ we denote the collection of neighbours of p in G_m by $V_{m,p}$. Lastly note that $N_m = \#V_m = 5^m \cdot 4 + 1$.

1.3 Hausdorff Dimensions

What is hausdorff dimension? We first have to look at Hausdorff measure.

We say the countable collection of set $\{U_i\}_{i\geq 1}$ is a *δ-cover* of K if $K \subset \bigcup_{i\geq 1} U_i$ and $\operatorname{diam} U_i \leq \delta$. Here $\operatorname{diam}(A) := \sup_{x,y\in A} d(x,y)$ where $d(\cdot,\cdot)$ is a metric.

Definition 1.4: Let X be a metric space. If $K \subset X$ and $d \in [0,\infty)$ we define

$$H^d_\delta(K) = \inf\left\{\sum_{i=1}^{\infty} \operatorname{diam}(U_i)^d \;:\; \bigcup_{i=1}^{\infty} U_i \supseteq K, \operatorname{diam}(U_i) < \delta\right\}.$$

Then the d-dimensional *Hausdorff measure* is given by

$$\mathcal{H}^d(K) = \lim_{\delta\to 0} H^d_\delta(K)$$

The Hausdorff measure defined above, is not a measure i the normal measure theory sense but ratherit is an outer measure. Though it is possible to prove that the Hausdorff measure restricted to Borel-set is a measure.

Moving swiftly along, we now want to look at the behaviour of $\mathcal{H}^s$ as s changes. So let $U_{i\,i\geq 1}$ is a δ-cover of K then we have for $0 \geq s < t$ that

$$\sum_{i\geq 1} \operatorname{diam}(U_i)^t = \sum_{i\geq 1} \operatorname{diam}(U_i)^{t-s} \operatorname{diam}(U_i)^s \leq \delta^{t-s} \sum_{i\geq 1} \operatorname{diam}(U_i)^s.$$

Taking infima on both sides we get that $H^t_\delta(K) \leq \delta^{t-s} H^s_\delta(K)$. If we let $\delta \to 0$ we see that if $\mathcal{H}^s(K) < \infty$ then $\mathcal{H}^s(K) = 0$ since $\delta^{t-s} \to 0$. Likewise if $\mathcal{H}^t > 0$ then $\mathcal{H}^s = \infty$. T his tells us that if there is a point, say κ, such $\mathcal{H}^\kappa$ is both positive and finite, then $\mathcal{H}^s$ will be 0 for all $s > \kappa$ and ∞ for all $s < \kappa$. That such value is called the *Hausdorff Dimension* of K. Formally written:

Definition 1.5: The *Hausdorff dimension*, denoted $d_h(K)$ is given by

$$d_h(K) = \inf\left\{s \geq 0 : \mathcal{H}^s(K) = 0\right\} = \sup\left\{s \geq 0 : \mathcal{H}^s(K) = \infty\right\}.$$

When there is no confusion with regard to what set we are working with, we denote the Hausdorff dimension by d_h.

Now it is quite difficult to determine the Hausdorff dimension from the definition. Luckily we can with a few assumption find the Hausdorff dimension quite easily. For that theorem we need an important lemma.

Lemma 1.6: Let μ be a measure such that $\mu(F) \in (0,\infty)$ and suppose there are numbers $c, \epsilon > 0$ such that for some $s \geq 0$

$$\mu(U) \leq c \operatorname{diam}(U)^s$$

for all sets U with $\operatorname{diam}(U) \leq \epsilon$. Then $\mathcal{H}^s(F) \geq \frac{\mu(F)}{c}$ and $s \leq d_h$

Proof. If $\{U_i\}_{i\in I}$ is any cover of F then

$$0 < \mu(F) \leq \mu\left(\bigcup_{i\in I} U_i\right) \leq \sum_{i\in I} \mu(U_i) \leq c \sum_{i\in I} \operatorname{diam}(U_i)^s$$

by definition of measures and properties of μ. Taking infimum we get $\mathcal{H}^s_\delta \geq \frac{\mu(F)}{c}$ and therefore $\mathcal{H}^s \geq \frac{\mu(F)}{c} > 0$ so by definition 1.5 $s \leq d_h$. □

Lemma 1.7: Let $\{V_i\}$ be a collection of disjoint open subset of $\mathbb{R}^n$ such that each V_i contains a ball of radius $a_1 r$ and is contained in a ball of radius $a_2 r$. Then any ball B of radius R intersects at most $(1+2a_2)^n a_1^n$ of the closures $\overline{V}_i$

Proof. By assumption we know that $\operatorname{diam}(V_i) \leq 2a_2 r$ for all V_i. So if some $\overline{V}_i$ intersect the ball with radius r, then $\overline{V}_i$ must be contained in the ball with radius $(2a_2+1)r$ concentric with the ball with radius r. Now let q denote the number of closures $\overline{V}_i$ that are intersecting with the ball with radius r. Then there exists q disjoint balls with radius $a_1 r$ that are contained in the ball with radius $(2a_2+1)r$, since by assumption each $\overline{V}_i$ contains a ball of radius $a_1 r$. It then follows that $q(a_1 r)^n \leq (2a_2+1)^n r^n$ which gives us the bound for q. □

We first define the open set condition

Definition 1.8: We say that a IFS $\mathcal{S}_i$ satisfy the *open set condition* if there exists a non-empty bounded open set V such that

$$V \supset \bigcup_{i=1}^{5} \mathcal{S}_i(V)$$

and each $\mathcal{S}_i(V)$ is disjoint.

Theorem 1.9: Suppose definition 1.8 holds for the similarities $\mathcal{S}_i$ on $\mathbb{R}^n$ with ratios $0 < c_i < 1$ for $1 \leq i \leq m$. If there for K holds

$$K = \bigcup_{i=1}^{m} \mathcal{S}_i(K) \tag{1.4}$$

then $d_h = s$ where s is given by

$$\sum_{i=1}^{m} c_i^s = 1. \tag{1.5}$$

Moreover, for this value of s, $0 < \mathcal{H}^s(K) < \infty$

Proof. Let $\mathcal{I}_k$ denote the set of all k-term sequences $(i_1, \dots i_k)$ with $1 \leq i_j \leq m$. For any set A and any sequence $(i_1, \dots i_k) \in \mathcal{I}_k$ we denote $A_{i_1,\dots i_k} = \mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_k}(A)$.

We will start by finding an upper bound for the Hausdorff measure. We do this by finding an appropriate δ-cover for K. It follows that by repeated use of (1.4) that

$$K = \bigcup_{i=1}^{m} \mathcal{S}_i(K) = \bigcup_{i=1}^{m} \mathcal{S}_i \left(\bigcup_{j=1}^{m} \mathcal{S}_j(K) \right) = \bigcup_{i,j=1}^{m} \mathcal{S}_i \circ \mathcal{S}_j(K) = \dots = \bigcup_{\mathcal{I}_k} \mathcal{S}_{i_1,\dots i_k}(K) = \bigcup_{\mathcal{I}_k} K_{i_1,\dots i_k}.$$

Now noting that $\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_k}$ has the ratios $c_{i_1} \dots c_{i_k}$, we see

$$\sum_{\mathcal{I}_k} \operatorname{diam}(K_{i_1,\dots i_k})^s = \sum_{\mathcal{I}_k} (c_{i_1} \dots c_{i_k})^s \operatorname{diam}(K)^s = \left(\sum_{i_1} c_{i_1}^s \right) \dots \left(\sum_{i_k} c_{i_k}^s \right) \operatorname{diam}(K)^s = \operatorname{diam}(K)^s$$

where the last equality comes from (1.5). Now for any $\delta > 0$ we can choose k so $\operatorname{diam}(K_{i_1,\dots i_k}) \left(\max_{1\leq i\leq m} c_i\right)^k \operatorname{diam}(K) \leq \delta$ since $\max_{1\leq i\leq m} c_i \in (0,1)$. Thus $\left\{K_{i_1,\dots i_k}\right\}_{\mathcal{I}_k}$ is a δ-cover for K thus $\mathcal{H}^s_\delta(K) \leq \operatorname{diam}(K)^s$ in which case $\mathcal{H}^s(K) \leq \operatorname{diam}(K)^s$ and we have an upper bound.

Next we need a lower bound for the Hausdorff dimension, which is substantially more difficult.

Let $\mathcal{I} = \left\{(i_1, i_2, \dots) : 1 \leq i_j \leq m\right\}$ be the set of all infinite sequences and let $\mathcal{I}_{i_1,\dots i_k} = \left\{(i_1, \dots, i_k, q_{k+1}, \dots) : 1 \leq q_j \leq m\right\}$, so the first initial k "terms" are chosen and then the rest are the remaining permutations.

Now we create an outer measure μ on subset of $\mathcal{I}$ defined by $\mu(\mathcal{I}_{i_1,\dots i_k}) = (c_{i_1} \dots c_{i_k})^s$. From this outer measure μ we construct an outer measure on K named $\tilde{\mu}$ defined in the following way. For some subset $A \subseteq K$ -

$$\tilde{\mu}(A) = \mu\left(\left\{(i_1, i_2, \dots) \in \mathcal{I} : x_{i_1, i_2, \dots} \in A \cap K\right\}\right)$$

Proofs that μ and $\tilde{\mu}$ are indeed outer measures can be found in [Haz22, Section 7].

Now we want to show that $\tilde{\mu}$ fulfills the properties of lemma 1.6.

Let V be the open set that satisfies definition 1.8. Since $\overline{V} \supset \mathcal{S}(\overline{V}) = \cup_{i=1}^m \mathcal{S}_i(\overline{V})$, then by theorem 1.3 the decreasing sequence $S^k(\overline{V})$ converges to K. In particularly we have $\overline{V} \supset S(\overline{V}) \supseteq \cup_{i=1}^m \mathcal{S}_i(\overline{V}) = F$ and $\overline{V}_{i_1,\dots i_k} = \mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_k}(\overline{V}) \supseteq \mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_k}(\overline{F}) = F_{i_1,\dots i_k}$ for each finite sequence $(i_1, \dots, i_k)$.

Now let B be any ball $r < 1$. We estimate $\tilde{\mu}(B)$ by considering sets $V_{i_1,\ldots i_k}$ with diameters comparable with that of B with closures intersecting $F \cap B$. Further we define $\mathcal{Q}$ as the set of sequences $(i_1, \ldots, i_k) \in \mathcal{I}$ such that i_k is the smallest k for which the following holds

$$\left(\min_{1\leq i\leq m} c_i\right) r \leq c_{i_1} \ldots c_{i_k} \leq r.$$

Note that no sequence in $\mathcal{Q}$ extends to another sequence. Since if two sequences are the same, but one is longer than the other the shorter sequence would be the one in $\mathcal{Q}$. Now by definition 1.8 $V_1, \ldots, V_m$ are disjoint and so are $V_{i_1,\ldots i_k,1}, \ldots, V_{i_1,\ldots i_k,m}$ for each $(i_1, \ldots, i_k)$. In particular we have that $\left\{V_{i_1,\ldots i_k} : (i_1, \ldots, i_k) \in \mathcal{Q}\right\}$ is a collection of disjoint sets. We further note that by (1.3) we have

$$K = \bigcup_{\mathcal{I}} \{x_{i_1,i_2,\ldots}\} \subseteq \bigcup_{\mathcal{Q}} K_{i_1,\ldots,i_k} \subseteq \bigcup_{\mathcal{Q}} \overline{V}_{i_1,\ldots,i_k}. \tag{1.6}$$

Next we choose a_1 and a_2 so that V contains a ball of radius a_1 and is contained in a ball of radius a_2. Then for all $(i_1, \ldots i_k) \in \mathcal{Q}$ the set $V_{i_1,\ldots i_k}$ contains a ball of radius $c_{i_1} \ldots c_{i_k} a_1$ and therefore one of radius $\left(\min_{1\leq i\leq m} c_i\right) a_1 r$. Further it is contained in a ball of radius $c_{i_1} \ldots c_{i_k} a_2$ and therefore one of radius $a_2 r$.

Now let $\mathcal{Q}_B$ be theset of all sequences $(i_1, \ldots, i_k) \in \mathcal{Q}$ such that $\overline{V}_{i_1,\ldots i_k}$ intersects B. By BALL LEMMA HERE there are at most

$$q = (1 + 2b)^n \left(\min_{1\leq i\leq m} c_i\right)^{-n} a^{-n}$$

sequences in $\mathcal{Q}_B$. Thus we find that if

$$x_{i_1,\ldots i_k} \in F \cap B \subseteq \bigcup_{\mathcal{Q}_B} \overline{V}_{i_1,\ldots i_k}, \quad \text{by (1.6)},$$

then there must be some k such that $(i_1, \ldots, i_k) \in \mathcal{Q}_B$ and therefore $(i_1, i_2, \ldots) \in \cup_{\mathcal{Q}_B} I_{i_1,\ldots i_k}$. We can thus conclude that

$$\tilde{\mu}(B) = \mu\left(\left\{(i_1, i_2, \ldots) \in \mathcal{I} : x_{i_1,i_2,\ldots} \in F \cap B\right\}\right) \leq \mu\left(\bigcup_{\mathcal{Q}_B} I_{i_1,\ldots,i_k}\right).$$

It then follows that

$$\tilde{\mu}(B) \leq \sum_{\mathcal{Q}_B} \mu\left(I_{i_1,\ldots,i_k}\right) = \sum_{\mathcal{Q}_B} \left(c_{i_1} \ldots c_{i_k}\right)^s \leq \sum_{\mathcal{Q}_B} r^s \leq q r^s.$$

Since any set U is contained in a ball of radius $\operatorname{diam}(U)$, we have that $\tilde{\mu} \leq q \operatorname{diam}(U)^s$. Thus by lemma 1.6 we have that $\mathcal{H}^s(F) \geq \frac{\tilde{\mu}(F)}{q} = q^{-1} > 0$, thus $0 < q^{-1} \leq \mathcal{H}^s(F) \leq \operatorname{diam}(F)^s < \infty$ thus by definition 1.5 we have that $s = d_h$ ☐

Spectral and Walk dimension

Further in this thesis we will wish to bound certain functions/fields/GRF? by a constant we call the spectral dimension. I depends on two other factors, namely the Hausdorff dimension, which we have just found, and of the walk dimension. Properly explaining the walk dimension is outside of the scope of this thesis, but we will give some intuition as to what it is.

Let B_t be a Brownian motion on KDont know where?, and $B(x, r)$ be the ball centered at x with radius r. then the *walk dimension* is the constant $d_w > 0$ such that

$$c_1 r^{d_w} \leq \mathbb{E}\left[\tau(x, r)\right] \leq c_2 r^{d_w}$$

where $\tau(x, r) = \inf\left\{t \geq 0 : B_t \notin B(x, r)\right\}$. The intuition behind the walk dimension is that it is a bound for when a Brownian motion leaves the fractal. This explanation for the walk dimension is a rather simplified and condensed one, for further reference see [Mar98, Chapter 3].

This then leads us to being able to define the spectral dimension as following:

Definition 1.10: Let d_h be the Hausdorff dimension and d_w the walk dimension then the spectral dimension is

$$d_s = \frac{2d_h}{d_W}$$

Chapter 2

And now for something completely different

In future section the Gaussian random fields will be defined using the eigenvalues and eigenfunctions of the Laplacian. We will therefore start by introducing first the discrete Laplacian, then the Kigami Laplacian and will culminate in the Dirichlet Laplacian.

We will now consider the discrete Laplacian on the fractal.

Definition 2.1: Let $\ell(V_m)$ be the set of functions of $f : V_m \to \mathbb{R}$. Then for any $f \in \ell(V_m)$ the *discrete Laplacian* on V_m is defined as

$$\Delta_m f(p) = \text{CONST} \sum_{q \in V_{m,p}} (f(q) - f(p)), \quad p \in V_m \backslash V_0$$

Remark 2.2: One is able to describe the discrete Laplacian as a matrix. First let $p_1, \dots p_{N_m}$ be the vertices in V_m then the *Laplacian matrix* is constructed in the following way:

$$L_{i,j} = \begin{cases} \#V_{m,p_i}, & \text{if } i = j \\ -1, & \text{if } i \neq j \text{ and } p_j \in V_{m,p_i} \\ 0, & \text{otherwise.} \end{cases}$$

Further $\#V_{m,p_i}$ is also called the *degree of* p_i

Now let $C(K)$ be the set of continuos functions on K. We then define

$$\mathcal{D} = \left\{ f \in C(K) \ : \ \exists g \in C(K), \lim_{m \to \infty} \max_{p \in V_m \backslash V_0} \left| \Delta_m f(p) - g(p) \right| = 0 \right\}$$

We will use the notation of $C_0(K)$ and D_0 to denote the sets of functions in $C(K)$ and D respectively which vanish on the boundary ∂K.

This set leads us to defining

Definition 2.3: The Kigami Laplacian $\Delta : \mathcal{D} \to C(K)$ is defined as

$$\Delta f(p) = g(p)$$

where g satisfies: $\lim_{m \to \infty} \max_{p \in V_m \backslash V_0} \left| \Delta_m f(p) - g(p) \right| = 0$

The definition above is one way of defining the Kigami Laplacian, but the domain doesn't lend itself to show what functions are actually contained in $\mathcal{D}$. We will therefore in this next section show another way to define the Kigami Laplacian with a more transparent domain and being able to define the Kigami Laplacian pointwise.

2.1 We're going to define it again

In this section we will follow arguments given by [Str06]. Much of the proof are going to rely on the standard graph Dirichlet form. Thus we are first going to define and prove some properties of this graph Dirichlet form. What to call these energies? Also the renormalized one

Definition 2.4: Given a finite connected graph G and it vertices V_m the for functions $u, v : V_m \to \mathbb{R}$ we define the *standard graph Dirichlet form* to be

$$E_m(u,v) = \sum_{x \sim y} \big(u(x) - u(y)\big)\,(v(x) - v(y)).$$

Notation: we write $E_m(u) = E_m(u,u)$.

Given some graph G and a function $u : V_m \to \mathbb{R}$, since we in the end want to work on the whole of the Vicsek set K, we now define an extension of u to V_m. This is done in the following way

Definition 2.5: Let K be a graph and $u : V_m \to \mathbb{R}$, then an *extension of* u is a function $u' : V_{m+n} \to \mathbb{R}$ where $u' \mid_{V_m} = u$.

It is obvious that $E_{m+n}(u') \geq E_m(u)$. Further one quickly notices that there must be a function that minimizes the graph Dirichlet form of a given graph. We denote such a function with $\tilde{u}$ and call it a *harmonic extension*.

Definition 2.6: Let $\tilde{u} : V_{m+n} \to \mathbb{R}$ with $\mid_{V_m} = u$ is called a *harmonic extension* if $E_{m+n}(\tilde{u}) \leq E_{m+n}(u')$ for all $u' : V_{m+n} \to \mathbb{R}$ with $u' \mid_{V_m} = u$.

In some cases we have that $E_{m+n}(\tilde{u}) = r^n E_m(u)$ for some $r \in (0,1)$. But instead of normalizing the values on E_m we will normalize E_{m+n}. Thus getting

$$r^{-n} E_{m+n}(\tilde{u}) = E_m(u), \quad \text{and} \quad r^{-n} E_{m+n}(u') \geq E_m(u)$$

for every extension u' of u. The constant r is called the *renormalization constant* From the equality above we can define the following

Definition 2.7: The *renormalized graph energies* are defined as

$$\mathcal{E}_m(u) = r^{-m} E_m(u)$$

Note that the sequence $\big\{\mathcal{E}_m(u)\big\}$ is a non-decreasing sequence.

Based on this definition we see that for a harmonic extension $\tilde{u}$ of u we have that

$$\mathcal{E}_{m+1}(\tilde{u}) = r^{-(m+1)} E_{m+1}(\tilde{u}) = r^{-m} E_m(u) = \mathcal{E}_m(u).$$

Extending this to the bilinear form we have for harmonic extension $\tilde{u}, \tilde{v}$ of u, v that

$$\mathcal{E}_{m+1}(\tilde{u}, \tilde{v}) = \mathcal{E}_m(u, v).$$

Actually one even has that for the bilinear form then if $\tilde{u}$ is the harmonic extension of u and v' is any extension of v then

$$\mathcal{E}_{m+1}(\tilde{u}, v') = \mathcal{E}_m(u, v). \tag{2.1}$$

See [Str06] for more information.

Now that we have 2.7, we can define what it means to be a harmonic function.

Definition 2.8: u is a *harmonic function* if u minimizes $\mathcal{E}_m(u)$ for all m, given boundary values V_0.

The harmonic functions will be used extensively throughout this section. We start by finding the renormalization constant for the Vicsek set.

Renormalization constant on the Vicsek set

I the previous section we have talked about a harmonic function, but without every constructing one. We will therefore in this next section show how to create an harmonic function for the Vicsek set for any start boundary values.

Given $V_0 = \{q_0, q_1, q_2, q_3, q_4\}$ and $u : V_0 \to \mathbb{R}$ then we have that

$$E_0(u) = \big(u(q_0) - u(q_1)\big)^2 + \big(u(q_0) - u(q_2)\big)^2 + \big(u(q_0) - u(q_3)\big)^2 + \big(u(q_0) - u(q_4)\big)^2.$$

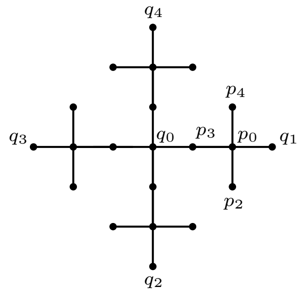

Diagram 2.1: Points on the Vicsek set

We are going to exploit symmetry of the Vicsek set. Since all four "limbs" of the Vicsek set are identical we will only look at one of the limbs since the other limbs will be the same computations. So if we want to minimize $E_1(u')$, we see for the limb between q_0 and q_1 that

$$E_1(u') = \big(u'(q_1) - u'(p_0)\big)^2 + \big(u'(p_0) - u'(p_2)\big)^2 \\ + \big(u'(p_0) - u'(p_3)\big)^2 + \big(u'(p_0) - u'(p_4)\big)^2 + \big(u'(p_3) - u'(q_0)\big)^2 .$$

We are then going to optimize with regards to $u'(p_i)$ for $i = 0, 1, 2, 4$. Taking the derivative of the right side and solving for 0 we get the following

$$\begin{aligned} 4u'(p_0) &= u'(q_1) + u'(p_2) + u'(p_3) + u'(p_4) \\ 2u'(p_3) &= u'(q_0) + u'(p_0) \\ u'(p_2) &= u'(p_0) \\ u'(p_4) &= u'(p_0). \end{aligned}$$

Solving these equations we get

$$\begin{aligned} u'(p_0) &= \frac{2}{3}u'(q_1) + \frac{1}{3}u'(q_0) \\ u'(p_2) &= \frac{2}{3}u'(q_1) + \frac{1}{3}u'(q_0) \\ u'(p_4) &= \frac{2}{3}u'(q_1) + \frac{1}{3}u'(q_0) \\ u'(p_3) &= \frac{2}{3}u'(q_0) + \frac{1}{3}u'(q_1). \end{aligned}$$

So what we have found is that to minimize the energy of the Vicsek set, then the energy of the next point is going to be 2/3 the energy from the closest point and 1/3 the energy from the next closest point.

If one calculates the energies then you will get

$$E_1(u') = \frac{1}{3}\sum_{i=1}^{4}(u'(q_i) - u'(q_0))^2 = \frac{1}{3}E_0(u')$$

Since the Vicsek set is a self-similar set, we have that going from V_1 to V_2 is the same as going from V_m to V_{m+1}. Thus we can just extend the above calculations to doing it m times, in which case

$$E_{m+1}(u') = \frac{1}{3}E_m(u')$$

Thus we get our renormalization constant for the vicsek set to be $r = 3$.

With the renormalization constant found we will return to finding the Kigmai Laplacian.

By definition 2.7 we have defined a sequence $\mathcal{E}_m$ for functions on V^*. It will therefore make sense to define the following limit

Definition 2.9: Let $u : V^* \to \mathbb{R}$ then define the limit

$$\mathcal{E}(u) = \lim_{m\to\infty} \mathcal{E}_m(u)$$

where we set

$$\operatorname{dom} \mathcal{E} = \left\{ u : \mathcal{E}(u) < \infty \right\}.$$

and denote $\operatorname{dom}_0 \mathcal{E}$ with the functions in $\operatorname{dom} \mathcal{E}$ that vanish V_0.

This definition only involves functions defined on V^*, but we find that functions in $\operatorname{dom} \mathcal{E}$ are uniformly continuos and therefore has an unique extension to K. We even have that $\operatorname{dom} \mathcal{E}$ is dense in $C(K)$.

Proposition 2.10:

(i) Let $u \in \operatorname{dom} \mathcal{E}$ then u is uniformly continuos on V^*.

(ii) $\operatorname{dom} \mathcal{E}$ is dense in $C(K)$.

Proof. We start with item i. Let $u \in \operatorname{dom} \mathcal{E}$ then $\mathcal{E}_m(u) \le \mathcal{E}(u)$ since $\mathcal{E}_m$ is non-decreasing in m. So for $x, y \in V_m$ if $y \in V_{m,x}$we have that

$$r^{-m}(u(x) - u(y))^2 \le \mathcal{E}_m(u) \le \mathcal{E}(u) \tag{2.2}$$

since $r^{-m}(u(x) - u(y))^2$ is a term in $\mathcal{E}_m(u)$. Thus

$$\left|u(x) - u(y)\right| \le r^{\frac{m}{2}} \mathcal{E}(u)^{\frac{1}{2}}.$$

Now consider a chain of points $x_m, x_{m+1}, \ldots, x_{m+k}$ where $x_{m+j} \in V_{m+j}$ and $x_{m+j} \in V_{m+j+1, x_{m+j+1}}$. Then we have

$$\left|u(x_m) - u(x_{m+k})\right| \le r^{\frac{m}{2}} \left(1 + r^{\frac{1}{2}} + \ldots + r^{\frac{k}{2}}\right) \mathcal{E}(u)^{\frac{1}{2}} \le \frac{r^m}{1 - r^{\frac{1}{2}}} \mathcal{E}^{\frac{1}{2}}$$

by using (2.2) and the triangle inequality multiple times. From the geometry of K we see that $x, y \in V^*$ belong to the same or adjacent m-cells then we can connect x to y by at most 2 such chains What does he mean?, so

$$\left|u(x) - u(y)\right| \le \frac{2r^{m/2}}{1 - r^{\frac{1}{2}}} \mathcal{E}(u)^{\frac{1}{2}}$$

and we have a bound for all $x, y \in V^*$.

For proof of item ii we refer to [Str06]. □

We are now going to state another way to define the Kigama Laplacian. We denote this Laplacian as Δ_μ since as we will see in the definition it depends on the measure μ.

Definition 2.11: Let $u \in \operatorname{dom} \mathcal{E}$ and let f be continuos, then $u \in \operatorname{dom}(\Delta_\mu)$ and $\Delta_\mu u = f$ if

$$\mathcal{E}(u, v) = \int_K f v \, \mathrm{d}\mu, \quad \forall v \in \operatorname{dom}_0 \mathcal{E}$$

While Δ_μ depends on the measure μ, we will still find that it coincides with the Kigami Laplacian Δ. Like with Δ we are going to want Δ_μ to be defined as the limit of a discrete operator. We are therefore going to again define a discrete graph Laplacian, the difference is that the following will be without constants.

Definition 2.12: We define the laplacian $\widetilde{\Delta}_m$ as

$$\widetilde{\Delta}_m = \sum_{y \in V_{m,x}} (f(y) - f(x))$$

Before moving on we will define a harmonic function which will be used extensively in the following proofs.

Definition 2.13: Given m and a point $x \in V_m \backslash V_0$ then define $\psi_x^{(m)}(y) = \delta_{xy}$ for $y \in V_m$.

I might seem that we have only defined $\psi_x^{(m)}$ on V_m, but actually it is defined on the whole of K. It simply follows that we create an harmonic extension of $\psi_x^{(m)}$.

The function $\phi_x^{(m)}$ can be used to describe $\widetilde{\Delta}_m$.

Proposition 2.14: Let $u \in \operatorname{dom} \mathcal{E}$ then

$$r^{-m}\widetilde{\Delta}_m u(x) = \int_K f\psi_x^{(m)} \, \mathrm{d}\mu$$

Proof. Since

$$\mathcal{E}_m(u, \psi_x^{(m)}) = r^{-m}\sum_{\substack{p \sim q \\ m}}(u(q) - u(p))(\psi_x^{(m)}(p) - \psi_x^{(m)}(q)) = r^{-m}\sum_{q\in V_{m,x}}(u(x) - u(q)) = r^{-m}\widetilde{\Delta}_m u(x)$$

where the second equality comes the fact that only edges containing x are going to see change in $\psi_x^{(m)}$. Now by (2.1) we have that

$$r^{-m}\widetilde{\Delta}_m u(x) = \mathcal{E}_m(u, \psi_x^{(m)}) = \mathcal{E}(u, \psi_x^{(m)}) = \int_K f\psi_x^{(m)} \, \mathrm{d}\mu$$

as desired. □

Before we begin the proof Δ_μ is the limit of $\widetilde{\Delta}_m$ we need some preliminary lemmas.

Lemma 2.15: Let $v : K \to \mathbb{R}$ be a harmonic function on the Vicsek set. If $v(x) = v(y_i) = 0$ for all $y_i \in V_{m,x}$ then $v(\omega) = 0$ for all $\omega \in \overline{B_x}$ where $B_x = \left(\bigcup_{k>0} S^k(x)\right) \cup \left(\bigcup_i \{y_i\}\right)$

Proof. It is clear that $v(\omega) = 0$ for any $\omega \in B_x$, since the energy of the points $F_w x$ with $|w| > 0$ depends on the energy of x and its neighbours.

So we need to check the points in $\overline{B_x}\backslash B_x$. Now since B_x is dense in $\overline{B_x}$, then for any $\omega \in \overline{B_x}$ we have that there exists a sequence $x_n \in B_x$ such that $x_n \to \omega$. By the fact that v is continuos on K it is continuos on $\overline{B_x}$, in which case $\left|v(x_n) - v(\omega)\right| \to 0$ as $n \to \infty$, but $v(x_n) = 0$ for all x_n, so $v(\omega) = 0$ as desired. □

What lemma 2.15 sates, is if any two points in V_m are neighbours, say x and y, and $v(x) = v(y) = 0$, then then function value at any point between x and y is also zero.

Lemma 2.16: Set $F_m = \operatorname{supp} \psi_x^{(m)}$, then

$$F_0 \supset F_1 \supset F_2 \supset F_3 \supset \dots$$

Further F_m is contained in the open ball $B(x, \epsilon_m)$ where $\epsilon_m \to 0$ as $m \to \infty$.

Proof. If we set $\Omega_m = \left\{z \in K : \psi_x^{(m)}(z) = \psi_x^{(m)}(y) = 0, \forall y \in V_{m,z}\right\}$ then $F_m = K\backslash \left(\bigcup_{\alpha\in\Omega_m} \overline{B_\alpha}\right)$. It is clear that $\Omega_m \subset \Omega_{m+1}$ and that there are new points in Ω_{m+1} that are not just a point between some already existing points $x_1, x_2 \in \Omega_m$. We therefore have from the way that F_m is constructed that $F_m \supset F_{m+1}$.

The containment of F_m in $B(x, \epsilon_m)$ follows from lemma 2.15 and that $F_m \supset F_{m+1}$. □

We are now ready to prove the pointwise formula

Theorem 2.17: Let $u \in \operatorname{dom} \Delta_\mu$. Then the pointwise formula

$$\Delta_\mu u(x) = \lim_{m\to\infty} r^{-m}\left(\int_K \psi_x^{(m)} \, \mathrm{d}\mu\right)^{-1}\widetilde{\Delta}_m u(x)$$

holds with the limit uniform across $V_*\backslash V_0$. Conversely, suppose u is a continuos function and the right side of the limit converges uniformly to a continuos function on $V_*\backslash V_0$. Then $u \in \operatorname{dom} \Delta_\mu$ and the equation holds.

Proof. Suppose $u \in \operatorname{dom} \Delta_\mu$ and $\Delta_\mu u = f$, then by proposition 2.14 we have that

$$r^{-m}\left(\int_K \psi_x^{(m)}\,\mathrm{d}\mu\right)^{-1}\widetilde{\Delta}_m u(x) = \frac{\int_K f\psi_x^{(m)}\,\mathrm{d}\mu}{\int_K \psi_x^{(m)}\,\mathrm{d}\mu}$$

for any $x \in V_m \backslash V_0$. Uniform converges of the right side is because of continuity of $f(x)$, so let $\epsilon > 0$ be given, then by lemma 2.16 we can choose m large enough so for all $x, y \in F_m$ that $\left|f(x) - f(y)\right| < \epsilon$, then

$$\begin{aligned}
\left|\frac{\int_K f\psi_x^{(m)}\,\mathrm{d}\mu}{\int_K \psi_x^{(m)}\,\mathrm{d}\mu} - f(x)\right| &= \left|\frac{\int_K f\psi_x^{(m)}\,\mathrm{d}\mu}{\int_K \psi_x^{(m)}\,\mathrm{d}\mu} - \frac{\int_K f(x)\psi_x^{(m)}\,\mathrm{d}\mu}{\int_K \psi_x^{(m)}\,\mathrm{d}\mu}\right| \\
&= \left|\frac{\int_K \big(f(y) - f(x)\big)\,\psi_x^{(m)}\,\mathrm{d}\mu}{\int_K \psi_x^{(m)}\,\mathrm{d}\mu}\right| \\
&\leq \frac{\int_{F_m} \left|f(y) - f(x)\right|\,\mathrm{d}\mu}{\int_K \psi_x^{(m)}\,\mathrm{d}\mu} \\
&\leq \frac{\epsilon \int_{F_m}\,\mathrm{d}\mu}{\int_K \psi_x^{(m)}\,\mathrm{d}\mu} \\
&= \epsilon
\end{aligned}$$

in which case

$$\lim_{m\to\infty} r^{-m}\left(\int_K \psi_x^{(m)}\,\mathrm{d}\mu\right)^{-1}\widetilde{\Delta}_m u(x) = \lim_{m\to\infty}\frac{\int_K f\psi_x^{(m)}\,\mathrm{d}\mu}{\int_K \psi_x^{(m)}\,\mathrm{d}\mu} = f(x) = \widetilde{\Delta}_\mu u(x)$$

Moving on to the second half of the proof. Suppose that $r^m\left(\int_K \psi_x^{(m)}\,\mathrm{d}\mu\right)^{-1}\widetilde{\Delta}_m u(x)$ converges uniformly to a continuos function f. Now for any $v \in \operatorname{dom}_0 \mathcal{E}$ we have

$$\mathcal{E}_m(u, v) = r^{-m}\sum_{\substack{x \sim y \\ m}} \big(u(y) - u(x)\big)\big(v(y) - v(x)\big).$$

If we collect all the terms that contain the factor $v(x)$ for a fixed $x \in V_m \backslash V_0$ then each $v(x)$ will be of the form $-v(x)\big(u(y) - u(x)\big)$ for each $y \underset{m}{\sim} x$. In which case we get

$$\begin{aligned}
\mathcal{E}_m(u, v) &= -r^{-m}\sum_{x \in V_m \backslash v_0} v(x)\left(\sum_{\substack{x \sim y \\ m}} \big(u(y) - u(x)\big)\right) \\
&= -\sum_{x \in V_m \backslash V_0} v(x) r^{-m}\widetilde{\Delta}_m u(x)
\end{aligned}$$

While it seems we lose many terms by removing $v(y)$, but they are instead contained in

$$\sum_{\substack{x \sim y \\ m}} \big(u(y) - u(x)\big).$$

Note that the sum is ”reset” for every x so we count both the relation $x \sim y$ and $y \sim x$.

Now if we define $f_m(x) = r^{-m}\left(\int_K \psi_x^{(m)}\,\mathrm{d}\mu\right)^{-1}\widetilde{\Delta}_m u(x)$ then $f_m \to f$ uniformly by assumption and we can rewrite the above as

$$\begin{aligned}
\mathcal{E}_m(u, v) &= -\sum_{x \in V_m \backslash V_0} v(x)\left(\int_K \psi_x^{(m)}\,\mathrm{d}\mu\right) r^{-m}\left(\int_K \psi_x^{(m)}\,\mathrm{d}\mu\right)^{-1}\widetilde{\Delta}_m u(x) \\
&= -\int_K \left(\sum_{x \in V_m \backslash V_0} v(x) f_m(x)\psi_x^{(m)}\right)\,\mathrm{d}\mu.
\end{aligned}$$

Following a quite lengthy argument, we have that

$$\mathcal{E}(u,v) = \lim_{m\to\infty} \mathcal{E}_m(u,v) = \lim_{m\to\infty} -\int_K \left(\sum_{x\in V_m\setminus V_0} v(x) f_m(x) \psi_x^{(m)} \right) \,\mathrm{d}\mu. = -\int_K v(y) f(y) \,\mathrm{d}\mu. \tag{2.3}$$

and we have that $u \in \mathrm{dom}\Delta_\mu$.

We therefore consider

$$\left| \sum_{x\in V_m\setminus V_0} v(x) f_m(x) \psi_x^{(m)}(y) - v(y) f(y) \right| \\ \leq \left| \sum_{x\in V_m\setminus V_0} \big(v(x) f_m(x) - v(y) f(y)\big)\, \psi_x^{(m)}(y) \right| + \left| \sum_{x\in V_m\setminus V_0} v(y) f(y) \psi_x^{(m)}(y) - v(y) f(y) \right|$$

Looking at each term individually we first see that

$$\left| \sum_{x\in V_m\setminus V_0} \big(v(x) f_m(x) - v(y) f(y)\big)\, \psi_x^{(m)} \right| \\ \leq \sum_{x\in V_m\setminus V_0} \Big(\big|v(x) f_m(x) - v(y) f_m(y)\big| + \big|v(y) f_m(y) - v(y) f(y)\big| \Big)\, \psi_x^{(m)}(y)$$

now again we will look at each term starting with $\big|v(y) f_m(y) - v(y) f(y)\big|$. Here

$$\big|v(y) f_m(y) - v(y) f(y)\big| = \big|v(y)\big| \big|f_m(y) - f(y)\big|$$

and since v is continuos on K, which is compact, there is a constant C_1 so $\big|v(y)\big| \leq C_1$ for all $x \in K$. Since f_m converges uniformly to f we can choose m_1 such that $\big|f_m(y) - f(y)\big| \leq \frac{\epsilon}{C_1}$ and the convergence of that term follows.

Next is $\big|v(x) f_m(x) - v(y) f_m(y)\big|$, here we find

$$\big|v(x) f_m(x) - v(y) f_m(y)\big| \leq \big|v(x)\big| \big|f_m(x) - f(y)\big| + \big|f_m(y)\big| \big|v(x) - v(y)\big|$$

First $\big|f_m(y)\big| \big|v(x) - v(y)\big|$, here $\big|f_m(y)\big| \leq C_2$ is uniformly bounded by some C_2. By lemma 2.16 and continuity of v we can choose m_2 so $\big|v(x) - v(y)\big| \leq \frac{\epsilon}{C_2}$ for all $x \in F_{m_2}$. The reason we can choose m_2 is because $\psi_x^{(m)}$ is multiplied onto all the terms so we just restrict v to F_{m_2}.

Second $\big|v(x)\big| \big|f_m(x) - f(y)\big|$, again $\big|v(x)\big| \leq C_1$, but

$$\big|f_m(x) - f(y)\big| \leq \big|f_m(x) - f(x)\big| + \big|f(x) - f_m(y)\big| \leq \big|f_m(x) - f(x)\big| + \big|f(x) - f(y)\big| + \big|f(y) - f_m(y)\big|$$

First choose m_3 so $\big|f_{m_3}(x) - f(x)\big| \leq \frac{\epsilon}{3C_1}$, next by continuity of f choose m_4 so $\big|f(x) - f(y)\big| \leq \frac{\epsilon}{3C_1}$ for all $x \in F_{m_4}$, lastly choose m_5 so $\big|f(y) - f_{m_5}(y)\big| \leq \frac{\epsilon}{3C_1}$.

So we now have for $m \geq \max\{m_1, m_2, \ldots, m_5\}$ that

$$\left| \sum_{x\in V_m\setminus V_0} \big(v(x) f_m(x) - v(y) f(y)\big)\, \psi_x^{(m)} \right| \leq \epsilon \sum_{x\in V_m\setminus V_0} \psi_x^{(m)} \leq \epsilon \sum_{x\in V_m\setminus V_0} 1 \leq \epsilon.$$

We are now going to look at $\left|\sum_{x\in V_m\setminus V_0} v(y) f(y) \psi_x^{(m)}(y) - v(y) f(y)\right|$ here

$$\left| \sum_{x\in V_m\setminus V_0} v(y) f(y) \psi_x^{(m)}(y) - v(y) f(y) \right| = \big|v(y) f(y)\big| \left| \sum_{x\in V_m\setminus V_0} \psi_x^{(m)}(y) - 1 \right|$$

Now to show $\left|\sum_{x\in V_m\setminus V_0} \psi_x^{(m)}(y) - 1\right| = 0$ as $m \to \infty$, we start by looking at $y \in V^*$. Choose m_6 so $y \in V_{m_6}$.

Next assume $y \in K$, then since V^* is dense in K we can choose a sequence $(x_n)_{n=1}$ where $x_n \to y$ as $n \to \infty$, and since $\psi_x^{(m)}$ is continuos on K then $\psi_x^{(m)}(x_n) \to \psi_x^{(m)}(y)$ as $n \to \infty$. This gives us the following

$$\left|\sum_{x \in V_m \backslash V_0} \psi_x^{(m)}(y) - 1\right| \leq \left|\sum_{x \in V_m \backslash V_0} \psi_x^{(m)}(y) - \sum_{x \in V_m \backslash V_0} \psi_x^{(m)}(x_n)\right| + \left|\sum_{x \in V_m \backslash V_0} \psi_x^{(m)}(x_n) - 1\right|$$

Now choose an n_1 so $\left|\psi_x^{(m)}(x_{n_1}) - \psi_x^{(m)}(y)\right| \leq \epsilon$ and an m_7 so $x_{n_1} \in V_{m_7}$. Both can then be bounded by $m \geq \max\{n_1, m_7\}$. Thus we have the convergence of (2.3). □

2.2 And back again

We notice that the definition of Δ and Δ_μ are the same. The definition for Δ does require us to take the max wrt. $p \in V_m \backslash V_0$, but since theorem 2.17 holds uniformly for any $p \in V^*\ V_0$ it does not matter which definition we choose.

We will now extend the Kigami Laplacian to the Dirichlet Laplacian.

We start by considering the following measure. where how and why does this measure come from? Why is it normalized hausdorff? The normalized Hausdorff measure on K which is given by the borel measure μ on K such for any $i_1, \ldots i_n \in \{1, \ldots, 5\}$

$$\mu\left(\mathcal{S}_{i_1} \circ \ldots \circ \mathcal{S}_{i_n}(K)\right) = 5^{-n}.$$

What does this mean? This measure is also d_h-Ahlfors regular, thus there exists $c_1, c_2 > 0$ such there for every $x \in K$ and $r \in [0, \mathrm{diam}(K)]$

$$c_1 r^{d_h} \leq \mu(B(x, r)) \leq c_2 r^{d_h}.$$

Next we have that we define the measure

$$\mu_m = \frac{1}{a_m} \sum_{p \in V_m} \delta_p \tag{2.4}$$

where $a_m = 5^m \cdot 4 + 1$.

This holds because $\mathcal{S}_i(K)$ are disjoint(except for single point both generate) for each i and each $\mathcal{S}_i$ "shrinks" the fractal down the that 1/5 of a quadrant. So $\sum_{x \in V_m} \delta_x(\mathcal{S}_{i_1,\ldots i_n}) = (1/5)^{-n} \cdot (5^m \cdot 4 + 1)$ since each $\mathcal{S}_i$ shrinks down the total number of points to 1/5 There is of course some relations between the measures we have just constructed.

Lemma 2.18: The sequence of measure $(\mu_m)_{m \geq 0}$ defined in (2.4) converges to the normalized Hausdorff measure μ on the Vicsek set K in the weak topology. That is

$$\lim_{m \to \infty} \int_K f \, \mathrm{d}\mu_m = \int_K f \, \mathrm{d}\mu$$

The proof of this follows from construction of the measures.

In the following we will follow [BC22] and [FS92] very closely in defining the Dirichlet Laplacian. Here we will encounter the concepts of *local regular Dirichlet form* and the *Friedrichs extension.* These concepts will not be explored in this thesis as they are only needed shortly to generate the Dirichlet Laplacian, thus they will be stated and used accordingly.

For any function $f : V_* \to \mathbb{R}$ we consider the quadratic form

$$\mathcal{E}_m(f, f) = \frac{const}{a_m} \sum_{\substack{x \sim y \\ m}} (f(x) - f(y))^2$$

the same as the one i already defined? but with constants. Define

$$\mathcal{E}(f, f) = \lim_{m \to \infty} \mathcal{E}_m(f, f)$$

and set

$$\mathcal{F}^0 = \{f \in C(K) : \mathcal{E}(f,f) < \infty, f = 0 \text{ on } \partial K\}$$

By theorem 4.1 we have that $(\mathcal{E}, \mathcal{F}_0)$ is a local Dirichlet form on $L^2(K,\mu)$ and lemma 4.1 in [FS92] we have that for any functions f, g on V_m vanishing on ∂K

$$\mathcal{E}_m(f,g) = -\int_{V_m^0} \Delta_m f(x) g(x) \, \mathrm{d}\mu_m(x)$$

and for $f \in \mathcal{D}_0$ and $g \in \mathcal{F}_0$ we have

$$\mathcal{E}(f,g) = -\int_K \Delta f(x) g(x) \, \mathrm{d}\mu.$$

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From theorem 4.2 in [FS92] the Friedrichs extension of the Kigami Laplacian Δ is the self-adjoint operator on $L^2(K,\mu)$ which is the generator of $(\mathcal{E}, \mathcal{F}_0)$. We still denote this generator Δ and this operator Δ with domain $\mathcal{D}(\Delta)$ is referred to as the Dirichlet Laplacian on K.

2.3 Discrete Fractional Gaussian Fields

Recalling remark 2.2 then we consider the eigenvalues $0 < \lambda_1^m \leq \lambda_2^m \leq \ldots \leq \lambda_{N_m}^m$ of $-\Delta_m$. Next let $(\Psi_i^m)_{1 \leq i \leq N_m}$ be the corresponding eigenfunctions with respect to the measure μ_m. This leads us to defining the following

Why are the eigenvalues greater than zero? and what is with the eigenfunctions wrt. measure?

Definition 2.19: Let $s \geq 0$ we define the *discrete Riesz kernel* on V_m as

$$G_s^m(x,y) = \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) \phi_i^m(y), \quad \text{for } x, y \in V_m$$

We can further by construction see that the matrix $(G_s^m(x,y))_{x,y \in V_m}$ is symmetric and is non-negative Why?, thus it can be the covariance matrix of a Gaussian vector.

Having constructed a discrete Laplacian we continue with constructing another Laplacian.

Definition 2.20: For $f \in \ell(V_m)$ and $s \geq 0$, the discrete fractional Laplacain $(-\Delta_m)^s : L^2(V_m, \mu_m) \to L^2(V_m, \mu_m)$ is defined by

$$(-\Delta_m)^s f(x) = \frac{1}{a_m} \sum_{y \in V_m} G_s^m(x,y) f(y) = \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) \frac{1}{a_m} \sum_{y \in V_m} \phi_i^m(y) f(y).$$

Proposition 2.21: $(-\Delta_m)^s$ is a uniformly bounded operator

Proof. Since

$$\begin{aligned}
\left\|(-\delta_m)^s f(x)\right\|_{L^2(V_m,\mu_m)} &\leq \left\| \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) \frac{1}{a_m} \sum_{y \in V_m} \phi_i^m(y) f(y). \right\|_{L^2(V_m,\mu_m)} \\
&\leq \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \frac{1}{a_m} \left\| \phi_i^m(x) \sum_{y \in V_m} \phi_i^m(y) f(y) \right\|_{L^2(V_m,\mu_m)} \\
&\leq (\lambda_1^m)^{-s} \|f\|_{L^2(V_m,\mu_m)}
\end{aligned}$$

where the last inequality comes from the fact that $0 < \lambda_1 < \lambda_2 < \ldots$. The uniform bound comes from the fact that $\inf_m \lambda_1^m > 0$ How? [FS92, lemma 5.2] ∎

Since we have now described the discrete fractional laplacian we will move on to the continuos fractional Laplacian.

Semi-groups

When stating the Fractional Laplacian one use that it admits a heat kernel through a semigroup generated by $-\Delta$. We will therefore first define a semigroup along with showing that $-\Delta$ generates a semigroup and lastly give some intuition to what the heat kernel is.

Definition 2.22: Let $\mathcal{H}$ be a Hilbert space. A strongly continuos self-adjoint contraction semigroup is a family of self-adjoint operators $(P_t)_{t\geq 0} : \mathcal{H} \to \mathcal{H}$

1. For $s, t \geq 0$, $P_t \circ P_s = P_{s+t}$ (semi-group property).
2. For every $f \in \mathcal{H}, \lim_{t\to 0} P_t f = f$ (strong continuity).
3. For every $f \in \mathcal{H}$ and $t \geq 0, \|P_t f\| \leq \|f\|$.

The next theorem also holds if we are given $(P_t)_{t\geq 0}$ as a strongly continuous self-adjoint contraction semigroup, then there exists a self-adjoint, non-positive and densly defined operator that generates $(P_t)_{t\geq 0}$, but we only need the the operator to genererate the semi-group.

Theorem 2.23: If A is a densely defined self adjoint operator on $\mathcal{H}$ then it is the generator of the strongly continuous self-adjoint contraction semigroup on $\mathcal{H}$ defined as $P_t = e^{tA}$.

Proof. We will simply show that e^{tA} fulfils definition 2.22. First

$$\begin{aligned} e^{tA}e^{sA} &= \sum_{n=0}^{\infty} \frac{(tA)^n}{n!} \sum_{n=0}^{\infty} \frac{(tA)^n}{n!} \\ &= \sum_{l=0}^{\infty} \sum_{n+m=l} \frac{t^n s^m A^l}{n!m!} \\ &= \sum_{l=0}^{\infty} \frac{(s+t)^l A^l}{l!} = e^{(s+t)A}. \end{aligned}$$

Next

$$\left| e^{tA} f - f \right| = \left| e^{tA} - 1 \right| |f| = 0$$

as $t \to 0$. And lastly since A is a non-positive operator then $e^{tA} < 1$ we find by the same calculations as above that

$$\left\| e^{tA} f - f \right\| = \left\| e^{tA} - 1 \right\| \|f\| \leq \|f\|$$

Maybe correct? □

Now giving some intuition for the heat kernel. If we look at the differential equations

$$\begin{cases} \frac{\mathrm{d}}{\mathrm{d}t} u + \Delta u = 0 \\ u(x, 0) = f(x) \end{cases}$$

where $u(x, t)$ and $f(x)$ are some continuous functions. Need help defining these... Now we see that $u(x, t) = e^{-t\Delta} f(x)$ is a solution to the differential equation. The heat kernel is then the continuous function $p_t(x, y)$ such that

$$e^{-t\Delta} f(x) = \int_K p_t(x, y) f(y) \, \mathrm{d}\mu(y).$$

Fractional Laplacian

Before we can define the continuos fractional Laplacian, we need to define the eigenvalues and eigenfunctions for $-\Delta$. A proof of the existence of the eigenvalues and eigenfunctions will not be given, but we will state the theorem.

The Laplacian Δ with domain $\mathcal{D}(\Delta)$ is by theorem 2.23 the generator of a strongly continuos Markov semigroup $\{P_t\}_{t\geq 0}$ on $L^2(K, \mu)$. This semigroup admits a heat kernel $p_t(x, y), t > 0, x, y \in K$, wrt. the

Hausdorff measure μ by Something above. It is called the Dirichlet heat kernel on K. Further this heat kernel satisfies for some $c_1, c_2 \in (0, \infty)$ find ref

$$p_t(x,y) \leq c_1 t^{-\frac{d_h}{d_w}} \exp\left(-c_2 \left(\frac{d(x,y)^{d_w}}{t}\right)^{\frac{1}{d_w-1}}\right) \tag{2.5}$$

for every $x, y \in K \times K$ and $t \in (0, \infty)$ whered_h is the Hausdorff dimension and d_w is the walk dimension.

We will actually find that the heat kernel can be described as a sum of eigenvalues and eigenfunctions. The following theorem both gives us these eigenvalues and functions.

Theorem 2.24: There exists a an orthonormal basis $\{\phi_i\}_{i=1}^{\infty}$ of $L^2(K,\mu)$ with ϕ_j being eigenfunctions of Δ. If λ_i is the eigenvalue belonging to ϕ_i, then $\lambda_i > 0$ and $\lim_{i\to\infty} \lambda_i = \infty$. Moreover

$$p_t(x,y) = \sum_{j=1}^{\infty} e^{-\lambda_j t}\phi_j(x)\phi_j(y)$$

where the convergence is uniform for all $t \in (0, \infty)$ and $x, y \in K \times K$

A detailed proof can be found in [Dod81, Theorem 2.2].

We actually have a result regarding the convergence of the eigenvalues.

Lemma 2.25:

$$\sum_{j=1}^{\infty} \frac{1}{\lambda_j^{2s}} < \infty. \iff s > \frac{d_h}{2d_w}$$

Proof. This follows from [Kig01, lemma 5.1.3] which states that there exists a constant $c_1, c_2 > 0$ such that

$$c_1 \cdot n^{2/d_s} \leq \lambda_n \leq c_2 \cdot n^{2/d_s} \tag{2.6}$$

Assume $s > \frac{d_h}{2d_w}$ then by (2.6)

$$\sum_{j=1}^{\infty} \frac{1}{\lambda_j^{2s}} \leq \sum_{j=1}^{\infty} \frac{1}{(c_1 n^{2/d_s})^{2s}} < \infty$$

since

$$\frac{4s}{d_s} > \frac{\frac{2d_h}{d_w}}{\frac{2d_h}{d_w}} = 1$$

the series is convergent.

Assume now that

$$\sum_{j=1}^{\infty} \frac{1}{\lambda_j^{2s}} < \infty.$$

then again by (2.6) we have that

$$\sum_{j=1}^{\infty} \frac{1}{(c_2 n^{2/d_s})^{2s}} \leq \sum_{j=1}^{\infty} \frac{1}{\lambda_j^{2s}} < \infty$$

and the left sum only converges if $4s/d_s > 1$ in which case $s > d_s/4 = d_h/2d_w$ as desired. ☐

Thus we are ready to define the fractional Laplacian.

Definition 2.26: Let $s \geq 0$, for $f \in L^2(K,\mu)$ the fraction Laplacian $(-\Delta)^{-s}$ is defined as

$$(-\Delta)^{-s} f = \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \int_K \phi_j(y) f(y)\, \mathrm{d}\mu(y)$$

Proposition 2.27:

(i) $(-\Delta)^{-s}$ is a linear operator

(ii) Let ϕ_n be an eigenfunction. Then $(-\Delta)^{-s}\phi_n = \frac{1}{\lambda_n^{-s}}\phi_n$

Proof. We start with the proof of (i) Let $f, g \in L^2(K,\mu)$ and α some constant, then by definition of $(-\Delta)^{-s}$ we have

$$\begin{aligned}(-\Delta)^{-s}(\alpha f + g) &= \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s}\phi_j \int_K \phi_j(y)(\alpha f(y) + g(y))\ \mathrm{d}\mu(y) \\ &= \alpha \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s}\phi_j \int_K \phi_j(y) f(y)\ \mathrm{d}\mu(y) + \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s}\phi_j \int_K \phi_j(y) g(y)\ \mathrm{d}\mu(y) \\ &= \alpha(-\Delta)^{-s} f + (-\Delta)^{-s} g\end{aligned}$$

Moving on to (ii). We note that since

$$\int_K \phi_j \phi_i\ \mathrm{d}\mu = \delta_{ij}$$

where δ_{ij} is the Kronecker delta. So for any eigenfunction ϕ_n we have

$$(-\Delta)^{-s}\phi_n = \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s}\phi_j(x) \int_K \phi_j(y)\phi_n/y)\ \mathrm{d}\mu(y) = \frac{1}{\lambda_n^{-s}}\phi_n.$$

□

Proposition 2.28: $(-\Delta)^{-s}$ is a bounded operator

Proof.

$$\begin{aligned}\left\|(-\Delta)^s f\right\| &= \left\| \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s}\phi_j \int_K \phi_j(y) f(y)\ \mathrm{d}\mu(y) \right\| \\ &\leq\end{aligned}$$

I don't know why since $\sum_{j\geq 0} \frac{1}{\lambda_i}^s$ is not convergent □

While the definition of the fractional Laplacian is not the most complicated, we are still able to rewrite it in another form.

We therefore define the fractional Reisz kernel Why is the definition different from other articles mainly $s > 0$ vs $s \geq 0$ and also $p_t + 1$

Definition 2.29: For parameter $s \geq 0$ we define the fractional Riesz kernel G_s by

$$G_s(x,y) = \frac{1}{\Gamma(s)} \int_0^{\infty} t^{s-1} p_t(x,y)\ \mathrm{d}t, \quad x, y \in K, x \neq y$$

where γ denotes the gamma function.

Before moving on we will state some important properties of G_s

Proposition 2.30: 1. If $s \in [0, d_h/d_w)$, there exists a constant $C > 0$ such that for every $x, y \in K, x \neq y$

$$G_s(x,y) \leq \frac{C}{d(x,y)^{d_h - s d_w}}$$

2. If $s = d_h/d_w$, there exists a constant $C > 0$ such that for every $x, y \in K, x \neq y$

$$G_s(x,y) \leq C\left|lnd(x,y)\right|$$

3. If $s > d_h/d_w$, there exists a constant $C > 0$ such that for every $x, y \in K, x \neq y$

$$G_s(x,y) \leq C$$

Before stating this proof we will quickly state a lemma needed

Lemma 2.31: There exists a constant $C > 0$ such that for every $x, y \in K$ and $t \geq 1$,

$$\left|p_t(x,y)\right| \leq Ce^{-\lambda_1 t}$$

where $\lambda_1 > 0$ is the first non-zero eigenvalue of $-\Delta$.

Proof. We have from theorem 2.24 that $p_t(x,y)$ has the expansion

$$p_t(x,y) = \sum_{j=1}^{\infty} e^{-\lambda_j t}\phi_j(x)\phi_j(y). \tag{2.7}$$

Since ϕ_j are eigenvalues, we have from definition of p_t? for any $t > 0$ that

$$\phi_j(x) = e^{\lambda_j t}\int_K p_t(x,y)\phi_j(y)\ \mathrm{d}\mu(y).$$

then from Cauchy-Schwarz inequality, we have for every $t > 0$

$$\left|\phi_j(x)\right| \leq e^{\lambda_j t}\left(\int_K p_t(x,y)^2\ \mathrm{d}\mu(y)\right)^{\frac{1}{2}}\left(\int_K \phi_j(y)^2\ \mathrm{d}\mu(y)\right)^{\frac{1}{2}} = e^{\lambda_j t}p_{2t}(x,x)^{1/2}$$

where the last equality comes from the fact that $\int_K \phi_i(y)\phi_j(y)\ \mathrm{d}\mu = \delta_{ij}$ so $\int_K \phi_j(y)^2\ \mathrm{d}\mu(y) = 1$. That combined with (2.7) gives us $\int_K p_t(x,y)^2\ \mathrm{d}\mu(y) = \int_K p_{2t}(x,y)\ \mathrm{d}\mu(y)$ which gives us the equality.

Now choosing $t = 1/4$ we get along with (2.5) that there is a constant $C > 0$ so for every $x \in K$ we have

$$\left|\phi_j\right| \leq Ce^{\lambda_j/4}.$$

Returning to (2.7) we ge get for every $x, y \in K$ and $t \geq 1$ that

$$\begin{aligned}
\left|p_t(x,y)\right| &\leq \sum_{j=1}^{\infty} e^{-\lambda_j t}\left|\phi_j(x)\right|\left|\phi_j(x)\right| \\
&\leq C^2\sum_{j=1}^{\infty} e^{-\lambda_j t}e^{\lambda_j/2} \\
&= C^2\sum_{j=1}^{\infty} e^{-\lambda_j t}e^{\lambda_j/2}e^{\lambda_j}e^{-\lambda_j} \\
&\leq C^2 e^{-\lambda_1 t}e^{\lambda_1}\sum_{j=1}^{\infty} e^{-\lambda_j/2}
\end{aligned}$$

here the last inequality comes from the fact that $0 < \lambda_1 < \lambda_2 < \ldots$ and $t \geq 1$ so

$$e^{\lambda_j(1-t)} \leq e^{\lambda_1(1-t)}.$$

Thus we have the desired bound. □

It now follows that

Proof of proposition 2.30. We have that

$$\begin{aligned}
G_s(x,y) &= \frac{1}{\Gamma(s)}\int_0^{\infty} t^{s-1}p_t(x,y)\ \mathrm{d}t \\
&= \frac{1}{\Gamma(s)}\int_0^{1} t^{s-1}p_t(x,y)\ \mathrm{d}t + \frac{1}{\Gamma(s)}\int_1^{\infty} t^{s-1}p_t(x,y).\ \mathrm{d}t
\end{aligned}$$

The intergral $\frac{1}{\Gamma(s)}\int_1^\infty t^{s-1}p_t(x,y)\ \mathrm{d}t$ can be uniformly bounded by a constant using lemma 2.31. Thus we only need to bound the integral $\int_0^1 t^{s-1}p_t(x,y)\ \mathrm{d}t$. By (2.5) we have that

$$\int_0^1 t^{s-1}p_t(x,y)\ \mathrm{d}t \le c_1\int_0^1 t^{s-1-\frac{d_h}{d_w}}\exp\left(-c_2\left(\frac{d(x,y)^{d_w}}{t}\right)^{\frac{1}{d_w-1}}\right).$$

Since the bound of the integral relies on the value of s we will split it up into the cases of the proposition.

Let $s > d_h/d_w$ then we simply have

$$\int_0^1 t^{s-1-\frac{d_h}{d_w}}\exp\left(-c_2\left(\frac{d(x,y)^{d_w}}{t}\right)^{\frac{1}{d_w-1}}\right) \le \int_0^1 t^{s-1-\frac{d_h}{d_w}}\ \mathrm{d}t.$$

Next if $s < d_h/d_w$, then using a change of variable $t = ud(x,y)^{d_w}$, we find that

$$\begin{aligned}
&\int_0^1 t^{s-1-\frac{d_h}{d_w}}\exp\left(-c_2\left(\frac{d(x,y)^{d_w}}{t}\right)^{\frac{1}{d_w-1}}\right).\\
&= d(x,y)^{sd_w-d_h}\int_0^{\frac{1}{d(x,y)^{d_w}}} u^{s-1-\frac{d_h}{d_w}}\exp\left(-c_2\left(\frac{1}{u}\right)^{\frac{1}{d_w+1}}\right)\ \mathrm{d}u\\
&\le d(x,y)^{sd_w-d_h}\int_0^{\infty} u^{s-1-\frac{d_h}{d_w}}\exp\left(-c_2\left(\frac{1}{u}\right)^{\frac{1}{d_w+1}}\right)\ \mathrm{d}u.
\end{aligned}$$

convergence of these integrals?. Lastly for $s = d_h/d_w$, we again use the change of variable $t = ud(x,y)^{d_w}$ and setting $R = \mathrm{diam}(K)$ and note that $0 < 1/R^{d_w} < 1/d(x,y)^{d_w}$, thus we find

$$\begin{aligned}
&\int_0^1 t^{s-1-\frac{d_h}{d_w}}\exp\left(-c_2\left(\frac{d(x,y)^{d_w}}{t}\right)^{\frac{1}{d_w-1}}\right).\\
&= \int_0^{\frac{1}{d(x,y)^{d_w}}} u^{-1}\exp\left(-c_2\left(\frac{1}{u}\right)^{\frac{1}{d_w+1}}\right)\ \mathrm{d}u\\
&\le \int_{\frac{1}{R^{d_w}}}^{\frac{1}{d(x,y)^{d_w}}} u^{-1}\exp\left(-c_2\left(\frac{1}{u}\right)^{\frac{1}{d_w+1}}\right)\ \mathrm{d}u + \int_0^{\frac{1}{R^{d_w}}} u^{-1}\exp\left(-c_2\left(\frac{1}{u}\right)^{\frac{1}{d_w+1}}\right)\ \mathrm{d}u\\
&\le \int_{\frac{1}{R^{d_w}}}^{\frac{1}{d(x,y)^{d_w}}} u^{-1}\ \mathrm{d}u + \int_0^{\frac{1}{R^{d_w}}} u^{-1}\exp\left(-c_2\left(\frac{1}{u}\right)^{\frac{1}{d_w+1}}\right)\ \mathrm{d}u\\
&\le d_w\left|\ln d(x,y)\right| + \int_0^{\frac{1}{R^{d_w}}} u^{-1}\exp\left(-c_2\left(\frac{1}{u}\right)^{\frac{1}{d_w+1}}\right)\ \mathrm{d}u\\
&\le C\left|\ln d(x,y)\right|.
\end{aligned}$$

Thus we have our desired bounds. ☐

2.4 The fractional Laplacian and its bounds

In the following we will prove some bounds for the fractional Laplacian. Since the goal of this thesis is to prove convergence of fractional gaussian fields within a specific space of test functions, we will only prove the lemmas for that space. The space of functions is given by

Definition 2.32: We consider the following space of test functions

$$S(K) = \left\{ f \in C_0(K) : \forall k \geq 0 \lim_{n\to\infty} n^k \left| \int_K \phi_n(y) f(y) \, \mathrm{d}\mu(y) \right| = 0 \right\}$$

We consider this space with the topology defined by the family of norms

$$\|f\|_k = \left\| (-\Delta)^{-k} f \right\|_{L^2(K,\mu)}.$$

By [Pie66, Theorem 2] and lemma 2.25 we have that $S(K)$ is a Frèchet Nuclear space. why? $D(T^\infty) = \cap_{n\geq 0} D(T^n)$ which doesn't look like our space. While the concepts of nuclear spaces is quite complicated, its properties will be very useful. It will allow us to prove convergence of Gaussian fields through the characteristics function and existence of Fractional Gaussian fields through a functional. Is this okay to write?

But first we will prove some lemmas regarding the fractional Laplacian and the fractional Reisz kernel.

Lemma 2.33: Let $s > 0$. For $f \in S(K)$ and $x \in K$

$$(-\Delta)^{-s} f = \int_K G_s(x,y) f(y) \, \mathrm{d}\mu(y)$$

Proof. We will first make sure the integral on the left side actually converges. But this holds because $S(K) \subset C(K)$ so f will be bounded on K. Thus we just need that for every $x \in K$ $\int_K G_s(x,y) \, \mathrm{d}\mu(y)$ is convergent. for $s \in [d_h/d_w, \infty)$ the convergence follows from proposition 2.30. Lastly for $s \in (0, d_h/d_w)$ then $d_h - sd_w < d_h$ we see that

$$\int_K G_s(x,y) \, \mathrm{d}\mu(y) \leq \int_K \frac{1}{d(x,y)^{d_h - sd_w}} \, \mathrm{d}\mu(y) < \infty$$

which follows from the Alfons regular hausdorff measure why?????.

Now using Fubini's theorem, theorem 2.24 and the definition of G_s we get

$$\begin{aligned}
\int_K G_s(x,y) f(y) \, \mathrm{d}\mu(y) &= \int_K \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} p_t(x,y) \, \mathrm{d}t f(y) \, \mathrm{d}\mu(y) \\
&= \int_K \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \sum_{j=1}^\infty e^{-\lambda_j t} \phi_j(x) \phi_j(y) \, \mathrm{d}t f(y) \, \mathrm{d}\mu(y) \\
&= \sum_{j=1}^\infty \phi_j(x) \int_K \phi_j(y) f(y) \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-\lambda_j t} \, \mathrm{d}t \, \mathrm{d}\mu(y) \\
&= \sum_{j=1}^\infty \frac{1}{\lambda_j^s} \phi_j(x) \int_K \phi_j(y) f(y) \, \mathrm{d}\mu(y) \\
&= (-\Delta)^{-s} f
\end{aligned}$$

where we use that $\frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-\lambda_j t} \, \mathrm{d}t = -\lambda_j^{-s}$ why? □

An immediate corollary follows:

Corollary 2.34: Let $s > 0$, for $f, g \in S(K)$

$$\int_K (-\Delta)^{-s} f (-\Delta)^{-s} g \, \mathrm{d}\mu = \int_K \int_K f(x) g(y) G_{2s}(x,y) \, \mathrm{d}\mu(x) \, \mathrm{d}\mu(y)$$

Proof. Now from the definition of $(-\Delta)^{-s}$, proposition 2.27 and the previous lemma, we find that

$$\begin{aligned}
\int_K (-\Delta)^{-s} f (-\Delta)^{-s} g \, \mathrm{d}\mu &= \langle (-\Delta)^{-s} f, (-\Delta)^{-s} g \rangle \\
&= \langle (-\Delta)^{-s} f, (-\Delta)^{-s} \sum_{j=1}^{\infty} \int_K \phi_j g \, \mathrm{d}\mu \phi_j \rangle \\
&= \langle (-\Delta)^{-s} f, \sum_{j=1}^{\infty} \frac{1}{\lambda_j^{-s}} \int_K \phi_j g \, \mathrm{d}\mu \phi_j \rangle \\
&= \langle f, (-\Delta)^{-s} \sum_{j=1}^{\infty} \frac{1}{\lambda_j^{-s}} \int_K \phi_j g \, \mathrm{d}\mu \phi_j \rangle \\
&= \langle f, \sum_{j=1}^{\infty} \frac{1}{\lambda_j^{-2s}} \int_K \phi_j g \, \mathrm{d}\mu \phi_j \rangle \\
&= \int_K f (-\Delta)^{-2s} g \, \mathrm{d}\mu \\
&= \int_K \int_K f(x) g(y) G_{2s}(x, y) \, \mathrm{d}\mu(x) \, \mathrm{d}\mu(y)
\end{aligned}$$

I use that $(-\Delta)^{-s}$ is a symmetric operator, but why? □

2.5 Discrete fractional Gaussian fields

Now that we have talked bout continuous laplacian we now turn to the discrete again.

Definition 2.35: Let $s \geq 0$, then a *discrete fractional Gaussian field*X_s^m with parameter s on V_m is a Gaussian vector indexed by V_m with mean zero and covariance matrix $G_2 s^m(x, y)$

For any $f \in \ell(V_m)$ we will use the notation

$$X_s^m(f) = \frac{1}{a_m} \sum_{p \in V_m} f(p) X_s^m(p)$$

We can then note that for $f, g \in \ell(V_m)$ we have that

$$\begin{aligned}
\mathbb{E}\left[X_s^m(f) X_s^m(g)\right] &= \frac{1}{2a_m} \sum_{p,q \in V_m} f(p) g(q) \mathbb{E}\left[X_s^m(p) X_s^m(q)\right] \\
&= \frac{1}{2a_m} \sum_{p,q \in V_m} f(p) g(q) G_{2s}^m(p, q) \\
&= \frac{1}{a_m} \sum_{p \in V_m} f(p) \left(\frac{1}{a_m} \sum_{q \in V_m} g(q) G_{2s}^m(p, q) \right) \\
&= \frac{1}{a_m} \sum_{p \in V_m} f(p) (-\Delta_m)^{-2s} g(p) \\
&= \frac{1}{a_m} \sum_{p \in V_m} (-\Delta_m)^{-s} f(p) (-\Delta_m)^{-s} g(p)
\end{aligned}$$

2.6 Fractional Gaussian Fields

Now that we have the fractional Laplacian let us state the fractional Gaussian field.

Definition 2.36: Let $s \geq 0$. A *fractional Gaussian field* X_s with parameter s on K is a centered Gaussian field $\{X_s(f) : f \in S(K)\}$ such that for any $f, g \in S(K)$

$$\mathbb{E}\left[X_s(f) X_s(g)\right] = \int_K (-\Delta)^{-s} f (-\Delta)^{-s} g \, \mathrm{d}\mu$$

We now quickly move to proving that the fractional Gaussian fields exist. First we will state the Bochner-Minlos theorem for nuclear spaces, since it is needed for the proof of existence. It will be given without proof but one can be found in [Joh03].

Theorem 2.37 (Bochner-Minlos)**:** Assume $\mathcal{X}$ is a nuclear space. Any continuos positive-definite linear functional Λ on $\mathcal{X}$ satisfying $\Lambda(0) = 1$ is the Fourier transform of a countably additive positive normalized measure on $\mathcal{X}'$

Lemma 2.38: The functional $\Phi : \mathcal{X} \to \mathbb{R}$ defined as $\Phi(v) = \exp(-\frac{1}{2}\langle v, v\rangle)$ is positive definite.

Proof. Let $v_1, \ldots v_n$ be elements of $\mathcal{X}$ and choose an orthonormal basis $e_1, \ldots e_m$ of span $\{v_1, \ldots, v_n\}$. Further let $Z = (Z_1, \ldots, Z_m)$ be a vector of independent standard normal real random variables, and note that for all $u \in \mathbb{R}^m$ we have that

$$\Phi\left(\sum_{j=1}^{\infty} u_j e_j\right) = \exp\left(-\frac{1}{2}\sum_{i=1}^{m} u_i^2\right) = \mathbb{E}\left[e^{\mathrm{i}u\cdot Z}\right].$$

How can $u \in \mathbb{R}^n$ be decomposed with another basis. This leads us to for $z \in ?^n$? where is z from? + definition of semi definite

$$z^T A^{\Phi} z = \sum_{j,k=1}^{n} z_j \overline{z_k} \Phi(v_j - v_k) = \sum_{j,k=1} z_j \overline{z_k} \mathbb{E}\left[e^{\mathrm{i}(v_j - v_k)\cdot Z}\right] = \mathbb{E}\left[\left|\sum_{j=1} z_j e^{iv_j \cdot Z}\right|^2\right] \geq 0$$

where does the last equality come from? □

Theorem 2.39: Let $s \geq 0$, there exists a centered Gaussian distribution X_s on $S'(K)$ such that for $f, g \in S(K)$

$$\mathbb{E}[X_s(f)X_s(g)] = \int_K (-\Delta)^{-s} f (-\Delta)^{-s} g \; \mathrm{d}\mu$$

Proof. The space $S(K)$ is a nuclear space why it is enough to use theorem 2.37(Bochner-Minlos) on the functional

$$\Lambda : f \to \exp\left(-\frac{1}{2}\|f\|_k^2\right).$$

This is enough since theorem 2.37 then gives an unique probability measure ν such

$$\exp\left(-\frac{1}{2}\|f\|_k^2\right) = \int_{S'(K)} e^{i\langle x, f\rangle} \; \mathrm{d}\nu(x).$$

Notice that the right side is an characteristic function for some random variable. If we choose $f = t \cdot g$ for some $g \in S(K)$ and $t \in \mathbb{R}$ Where is t from?(notice that $t \cdot g \in S(K)$) then the left side of the equation becomes the characteristic function for the normal distribution with mean zero and variance $\|f\|_s^2$. Thus we have the X_s we wanted.

Now to prove that Λ is satisfies the requirements of theorem 2.37. Obviously $\Lambda(0) = 1$, further it follows from the fact that $\|t \cdot f\|_k^2$ is an inner product that withlemma 2.38 Λ is positive definite. Lastly we need to prove it is continuous at zero.

First we note that since $(-\Delta)^{-s}$ is a bounded operator we have that there exists some constant $C > 0$ so $\left\|(-\Delta)^{-s} f\right\|_{L^2(K,\mu)} \leq C\|f\|_{L^2(K,\mu)}$.

Now let $\epsilon > 0$ be given, choose

$$\delta = \begin{cases} \left(\frac{-2\cdot\ln(1-\epsilon)}{C}\right)^{\frac{1}{2}}, & \text{when } \epsilon \in (0,1) \\ 1, & \text{when } \epsilon \geq 1. \end{cases}$$

First if $\epsilon \geq 1$ notice that

$$\left|\exp\left(-\frac{1}{2}\|f\|_s^2\right) - \exp\left(-\frac{1}{2}\|0\|_s^2\right)\right| \leq 1 \leq \epsilon, \quad \forall f \in S(K)$$

in which we can choose δ to be any real number larger than 0. Moving on to $\epsilon \in (0,1)$ we get

$$\begin{aligned}
\left|\exp\left(-\frac{1}{2}\|f\|_s^2\right) - \exp\left(-\frac{1}{2}\|0\|_s^2\right)\right| &\leq \left|\exp\left(-\frac{1}{2}C\|f\|_{L^2(K,\mu)}\right) - 1\right| \\
&= \left|\exp\left(-\frac{1}{2}C\int_K |f|^2 \, \mathrm{d}\mu\right) - 1\right| \\
&\leq \left|\exp\left(-\frac{1}{2}C\int_K \delta^2 \, \mathrm{d}\mu\right) - 1\right| \\
&= \left|\exp\left(-\frac{1}{2}C\int_K \frac{-2\cdot\ln(1-\epsilon)}{C} \, \mathrm{d}\mu\right) - 1\right| \\
&= \left|\exp\left(\ln(1-\epsilon)\right) - 1\right| \\
&= \epsilon
\end{aligned}$$

in the third to last equality using that μ is a probability measure. In which case we have continuity in 0 as desired. □

In the following we want to prove that a FGF has an L^2 density What is it? if and only if $s > \frac{d_h}{2d_w}$. But for this we need to construct white noise.

Let $\mathcal{K}$ be the Borel σ-algebra on K, then we can define

Let W the measure which is uniquely defined by the char. function

$$\int_{S'(K)} \exp(i\langle\omega,\phi\rangle) \, \mathrm{d}W(\omega) = \exp\left(-\frac{1}{2}\|\phi\|_{L^2}^2\right)$$

for $\phi \in S(K)$. This measure again comes from theorem 2.37. I dunno if this is correct

This will be done following [Le 16].

This whole section then since μ is a finite measure and therefore σ- finite. It then follows from [Le 16, Prop. 1.13] that there exists a probability field $(\Omega, \mathcal{F}, \mathbb{P})$ with a real-valued centered Gaussian random measure $W : L^2(\Omega, \mathcal{F}, \mathbb{P}) \to \mathcal{K}$ with density/intesity? μ.

If we now write $W(A) = W$

Definition 2.40: Let $s > \frac{d_h}{2d_w}$. The fractional Brownian field with parameter s is defined as

$$\widetilde{X}_s(x) = \int_K G_s(x,y) W(\,\mathrm{d}y)$$

Note that since $s > \frac{d_h}{2d_w}$ we can write

$$\widetilde{X}_s(x) = \sum_{i=1}^{\infty} \lambda_i^{-s}\phi_i(x) W_i$$

where$W_i = W(\phi_i)$ is an i.i.d sequence of Gaussian random variables with mean zero and variance one.

We will now show that one can construct a fractional Gaussian Field from the fractional Brownian field

Proposition 2.41: Let $s > \frac{d_h}{2d_w}$. Then the Gaussian random field defined by

$$X_s(f) = \int_K f(x)\widetilde{X}_s(x) \, \mathrm{d}\mu(x), \quad f \in S(K)$$

has the law of a fractional Gaussian field with parameter s. Moreover, if there exists a Gaussian field $\left(\widetilde{X}_s(x)\right)_{x\in K}$ on K with

$$\mathbb{E}\left[\int_K \widetilde{Y}_S(x)^2 \, \mathrm{d}\mu(x)\right] < \infty$$

such that the Gaussian field defined by

$$Y_s(f) = \int_K f(x)\widetilde{Y}_s(x)\ \mathrm{d}\mu(x), \quad f \in S(K)$$

has the law of fractional Gaussian field with parameter s, then $s > \frac{d_h}{2d_w}$

Proof. Assume $s > \frac{d_h}{2d_w}$. The from Fubini's theorem we have for every $f \in S(K)$ that a.s

$$\int_K f(x)\widetilde{X}_s(x)\ \mathrm{d}\mu(x) = \int_K f(x)\int_K G_s(x,y)W(\ \mathrm{d}y)\ \mathrm{d}\mu(x) = \int_K (-\Delta)^{-s}f(y)W(\ \mathrm{d}y)$$

in which case by definition of W we have that $X_s(f) = \int_K f(x)\widetilde{X}_s(x)\ \mathrm{d}\mu(x)$ is a Gaussian random variable with mean zero and variance

$$\int_K \left|(-\Delta)^{-s}f(x)\right|^2\ \mathrm{d}\mu(x).$$

and thus a fractional Gaussian field with parameter s.

Now assume that there is a Gaussian field $\left(\widetilde{X}_s(x)\right)_{x\in K}$ on K with

$$\mathbb{E}\left[\int_K \widetilde{Y}_S(x)^2\ \mathrm{d}\mu(x)\right] < \infty$$

such that the Gaussian field defined by

$$Y_s(f) = \int_K f(x)\widetilde{Y}_s(x)\ \mathrm{d}\mu(x), \quad f \in S(K)$$

has the law of fractional Gaussian field with parameter s. Then by spectral decomposition reference? we have

$$\widetilde{Y}_s(x) = \sum_{i=1}^{\infty} Y_s(\phi_i)\phi_i(x).$$

Note that $(Y(\phi_i))_{i\geq 1}$ is a sequence of independent Gaussian random variables with mean zero why mean zero? and why independent Gaussian? and variance $()\lambda_i^s)_{i\geq 1}$. By definition 2.36 and proposition 2.27 we have that

$$\mathbb{E}\left[Y_s(\phi_i)Y_s(\phi_j)\right] = \int_K (-\Delta)^{-s}\phi_i(-\Delta)^{-s}\phi_j\ \mathrm{d}\mu = \frac{1}{(\lambda_j\lambda_i)^s}\int_K \phi_i\phi_j\ \mathrm{d}\mu = \frac{1}{(\lambda_j\lambda_i)^s}\delta_{ij}.$$

Since $\mathbb{E}\left[\int_K \widetilde{Y}_s(x)^2\ \mathrm{d}\mu(x)\right] < \infty$ we have that

$$\begin{aligned}
\mathbb{E}\left[\int_K \widetilde{Y}_s(x)^2\ \mathrm{d}\mu(x)\right] &= \mathbb{E}\left[\int_K \left(\sum_{i=1}^{\infty} Y_s(\phi_i)\phi_i(x)\right)^2\ \mathrm{d}\mu(x)\right] \\
&= \mathbb{E}\left[\sum_{i=1}^{\infty} Y_s(\phi_i)^2\int_K \phi_i(x)\phi_i(x)\ \mathrm{d}\mu(x)\right] \\
&= \sum_{i=1}^{\infty}\mathbb{E}\left[Y_s(\phi_i)^2\right] \\
&= \sum_{i=1}^{\infty}\frac{1}{\lambda_i^{-2s}} < \infty
\end{aligned}$$

where we use that $\int_K(\sum_{i=1}^{\infty} Y_s(\phi_i)\phi_i(x))^2\ \mathrm{d}\mu(x) = \int_K \sum_{i=1}^{\infty} Y_s(\phi_i)^2\phi_i(x)^2\ \mathrm{d}\mu(x)$ which stems from the fact that $\int_K \phi_i\phi_j\ \mathrm{d}\mu = \delta_{ij}$ and $Y_s(\phi_i)$ is a constant wrt. $\mu(x)$.

We now have that

$$\mathbb{E}\left[\int_K \widetilde{Y}_s(x)^2\ \mathrm{d}\mu(x)\right] = \sum_{i=1}^{\infty}\frac{1}{\lambda_i^{-2s}} < \infty$$

in which case by lemma 2.25 we have that $s > \frac{d_h}{2d_w}$ ∎

Chapter 3

Main Theorem

Now that we have presented and proved the existence of fractional Gaussian fields X_s on K and discrete fractional Gaussian fields X_s^m on V_m out next task is proving they converge. Therefore in this section our main goal is to show that X_s^m converges in distribution to X_s.

3.1 Preliminary lemmas

We have now proven the existence of the Gaussian fields and are now ready to prove the convergence.

Lemma 3.1: For all $f \in S(K)$ and $s \geq 0$ the following holds

$$\frac{1}{a_m} \sum_{p \in V_m} f_m(p)(-\Delta_m)^{-2s} f_m(p) \to \int_K f(x)(-\Delta)^{-2s} f(x) \, \mathrm{d}\mu(x), \quad \text{as } m \to \infty$$

Proof. For $s = 0$ we find that

$$\begin{aligned}
(-\Delta_m)^{-0} f &= \sum_{i=1}^{N_m} \frac{1}{(\lambda_i^m)^0} \phi_i^m(x) \frac{1}{a_m} \sum_{y \in V_m} \psi_i^m(y) f(y) \\
&= \sum_{i=1}^{N_m} \phi_i^m(x) \int_{V_m} \phi_i^m(y) f(y) \, \mathrm{d}\mu_m \\
&= f_m.
\end{aligned}$$

Likewise by definition of $(-\Delta)^{-s}$ we find

$$(-\Delta)^{-0} f = \sum_{i=1}^{\infty} \phi_i \int_K \phi_i(y) f(y) \, \mathrm{d}\mu = f.$$

and the lemma follows from REF!!!. Now assume $s > 0$ then by Fubini's theorem and Some random ass fact we have that

$$\frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p)(-\Delta_m)^{-2s} f_m(p) = \frac{1}{\Gamma(2s)} \int_0^\infty t^{2s-1} \frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p).$$

Now by B.2.7 in [Kig01] and the definition of f_m we find that INSERT CALCULATION HERE

$$\begin{aligned}
\frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p) &= \frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m \sum_{j=1}^{\infty} \langle f_m, \phi_i \rangle \phi_i \\
&= \frac{1}{a_m} \sum_{p \in V_m} f_m(p) \sum_{j=1}^{\infty} \langle f_m, \phi_j \rangle P_t^m \phi_j \\
&= \frac{1}{a_m} \sum_{p \in V_m} f_m(p) \sum_{j=1}^{\infty} \langle f_m, \phi_j \rangle e^{-\lambda_j^m t} \phi_j \\
&\leq e^{-\lambda_1^m t} \frac{1}{a_m} \sum_{p \in V_m} f_m(p)^2
\end{aligned}$$

where we can guarantee the inequality for all j since $0 < \lambda_1 < \lambda_2 < .\lambda_3 < ...$

Note that $\sup_m \frac{1}{a_m} \sum_{p \in V_m} f_m(p)^2 < \infty$ since $f \in S(K) \subset C(K)$ so f is bounded and $\#V_m < \infty$ for all m. We can now by dominated convergence, Fubini's theorem and lemma 3.2 find that

$$\begin{aligned}\lim_{m\to\infty} \frac{1}{a_m} \sum_{p \in V_m} f_m(p)(-\Delta_m)^{-2s} f_m(p) &= \frac{1}{\Gamma(2s)} \int_0^\infty t^{2s-1} t^{2s-1} \lim_{m\to\infty} \frac{1}{a_m} \sum_{p\in V_m} f_m(p) P_t^m f_m(p) \ \mathrm{d}t \\ &= \frac{1}{\Gamma(2s)} \int_0^\infty t^{2s-1} \int_K f(x) P_t f(x) \ \mathrm{d}\mu(x) \ \mathrm{d}t \\ &= \frac{1}{\Gamma(2s)} \int_K f(x) \int_0^\infty t^{2s-1} P_t f(x) \ \mathrm{d}t \ \mathrm{d}\mu(x) \\ &= \int_K f(x)(-\Delta)^{-2s} f(x) \ \mathrm{d}\mu(x)\end{aligned}$$

as desired. □

In the following proof we will use results from [Kur69]. Before stating the lemma and proof we will give some arguments as to why we are allowed to use these results.

Firstly we note that since Δ is a bounded operator the by [Paz92, Theorem 1.2] it is the infinitesimal generator of a uniformly continuos semigroup.

Secondly we have that the sequence of laplacians $\{\Delta_m\}_{m\geq 0}$ indeed coincides with extended limit defined in [Kur69, page 355]. This follows from theorem 2.17.

Lastly the range $\mathcal{R}(\lambda_0 - \Delta)$ is dense in L^2 why?

Lemma 3.2: For all $f \in S(K)$ and $t > 0$,

$$\lim_{m\to\infty} \frac{1}{a_m} \sum_{p\in V_m} f_m(p) P_t^m f_m(p) = \int_K f(x) P_t f(x) \ \mathrm{d}\mu(x)$$

where $f_m = f \mid_{V_m}$.

Proof. In this proof we wish to use [Kur69, theorem 2.1] so recall that by the definition of the Laplacian on K we have that

$$\lim_{m\to\infty} \sup_{p\in V_m} \left|\Delta_m f_m(p) - \Delta f(p)\right| .$$

Thus by [Kur69, theorem 2.1] we have that for all $t \geq 0$

$$\lim_{m\to\infty} \sup_{p\in V_m} \left|P_t^m f_m(p) - P_t f(p)\right| = 0. \tag{3.1}$$

So if we now write

$$\frac{1}{a_m} \sum_{p\in V_m} f_m(p) P_t^m f_m(p) = \int_{V_m} f_m P_t^m f_m \ \mathrm{d}\mu_m.$$

Thus

$$\int_{V_m} f_m P_t^m f_m \ \mathrm{d}\mu_m = \int_{V_m} f_m \left(P_t^m f_m - (P_t f)_m\right) \ \mathrm{d}\mu_m + \int_{V_m} f_m (P_t f)_m \ \mathrm{d}\mu_m.$$

The first integral converges to 0 as $m \to \infty$ because of (3.1). Further by lemma we ave $\int_{V_m} f_m(P_t f)_m \ \mathrm{d}\mu_m \to \int_K f P_t f \ \mathrm{d}\mu$ as $m \to \infty$. □

3.2 Main Theorem

Theorem 3.3: Let $s \geq 0$, then X_s^m converges to X_s in distribution in $S'(K)$ when $m \to \infty$

Proof. First since $S(K)$ is a nuclear space, by [BDW17, Corollary 2.4] it is enough to prove the convergence of the characteristic functional, e.i that for all $f \in S(K)$

$$\mathbb{E}\left[\exp(itX_s^m(f_m))\right] \to \mathbb{E}\left[\exp(itX_s(f))\right], \quad \text{as } m \to \infty$$

where $f_m = f\mid_{V_m}$. It follows fra INSERT REF that

$$\mathbb{E}\left[\exp(itX_s^m(f_m))\right] = \exp\left(-\frac{1}{2}t^2\mathbb{E}\left[(X_s^m(f_m))^2\right]\right) = \exp\left(-\frac{1}{2a_m}t^2\sum_{p\in V_m} f_m(p)(-\Delta_m)^{-2s}f_m(p)\right)$$

Likewise we find that via coll. 2.11 + 2.9 we get that

$$\mathbb{E}\left[\exp(itX_s(f))\right] = \exp\left(-\frac{1}{2}t^2\mathbb{E}\left[(X_s(f))^2\right]\right) = \exp\left(-\frac{1}{2}t^2\int_K f(-\Delta)^{-2s}f\ \mathrm{d}\mu\right).$$

Where we notice that

$$\begin{aligned}\mathbb{E}\left[(X_s(f))^2\right] &= \int_K (-\Delta)^s f(-\Delta)^s f\ \mathrm{d}\mu \\ &= \int_K\int_K f(p)f(q)G_{2s}(p,q)\ \mathrm{d}\mu(p)\ \mathrm{d}\mu(q) \\ &= \int_K f(-\Delta)^{-2s}f\ \mathrm{d}\mu.\end{aligned}$$

The proof now follows from lemma 3.1. □

3.3 Simulations

I have some pretty pictures. Want to se the fields $X_s^m(x) = \sum \lambda_j\phi_j$ to $\widetilde{X}_s$?

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